

Team Number: \_\_\_\_\_ School & Team Name: \_\_\_\_\_

Student Names: \_\_\_\_\_

### PLANE A PLANE B ← CHECK-IN (UP TO 2 AIRPLANES)

#### Construction Parameters:

|   |   |   |   |       |                                                                                                                                                                                                          |
|---|---|---|---|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T | F | T | F | 3.a.  | If kits are used, they do not contain any pre-glued joints or pre-covered surfaces.                                                                                                                      |
| T | F | T | F | 3.b.  | Boron filaments are not used in the construction of the airplanes or boxes.                                                                                                                              |
| T | F | T | F | 3.c.  | The airplane in its flight configuration fits fully into a lidless, team-provided Measurement box with external dimensions 32.0 cm x 24.0 cm x 47.0 cm, including any external protuberances on the box. |
| T | F | T | F | 3.d.  | No change in chord, span, length, or total lifting area occurs after removing the airplane from its box and throughout the flight itself.                                                                |
| T | F | T | F | 3.e.  | Total mass of the airplane, excluding the rubber motor(s), is 8.0 g or more.                                                                                                                             |
| T | F | T | F | 3.f.  | Variable-pitch propellers that include mechanisms that actively change the propeller diameter or blade angle are not used.                                                                               |
| T | F | T | F | 3.g.  | All airplane lifting forces are generated by wing(s).                                                                                                                                                    |
| T | F | T | F | 3.h.  | The sole power for the airplane come from the rubber motor(s) that remain attached to the airplane throughout the flight.                                                                                |
| T | F | T | F | 3.h.i | The total mass of the rubber motor or rubber motor set, including attachments such as O-rings, does not exceed 1.5 g.                                                                                    |
| T | F | T | F | 3.i.  | Participants are able to answer questions on design, construction, and operation of the device per the Building Policy found on <a href="http://www.soinc.org">www.soinc.org</a> .                       |

1. T F 2. T F ALL CONSTRUCTION PARAMETERS ABOVE ARE MET. IF F, Tier 2.

#### FLIGHT LOG

3. Is the Flight Log complete, partial, or not present?

**Complete** (1.2x bonus multiplier)

**Partial** (1.1x bonus multiplier)

**Not present**

5.a. *Flight Log includes 6 or more parameters (motor size before wind-up; number of turns on launch or torque at launch; flight time; and 3 more chosen by the team) with data for 10 or more test flights prior to the competition.*

5.b. *All numeric data includes units. Torque units may be arbitrary.*

#### FLIGHT PERIOD (10 MINUTES)

##### Competition Parameters:

|   |   |         |                                                                                                                                                                                                            |
|---|---|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T | F | 4.b.i.3 | No practice flights are flown in the final half-hour of the event's last period. (Except for trim flights during a team's 10-minute Flight Period)                                                         |
| T | F | 4.b.ii  | Participants do not receive outside materials, assistance, or communication once they enter the cordoned off competition area to trim, practice, or compete (Except as permitted by the Event Supervisor). |
| T | F | 4.e.ix. | Participants do not steer the helicopter during flight.                                                                                                                                                    |
| T | F | 4.e.x.  | Students are on the floor to launch and do not use artificial aids to increase launch height.                                                                                                              |

4. T F ALL COMPETITION PARAMETERS ABOVE ARE MET. If F, Tier 2.

#### 1<sup>ST</sup> FLIGHT

5. A B

#### 2<sup>ND</sup> FLIGHT

7. A B

#### (OFFICIAL FLIGHTS)

Which plane from above was used?

6. \_\_\_\_:\_\_\_\_.\_\_\_\_

8. \_\_\_\_:\_\_\_\_.\_\_\_\_

← Median Flight Time of 3 timers  
(minutes:seconds)

9. T F General Rule: The team is disqualified. (Notify the team and their coach as soon as possible.)